

Ch. 7 MOSFET Technology Scaling, Leakage Current, and Other Topics

MOS ICs have met the world's growing needs for electronic devices for computing, communication, entertainment, automotive, and other applications with steady improvements in cost, speed, and power consumption. Such steady improvements in turn stimulate and enable new applications and fuel the growth of IC sales. There is now an entrenched expectation that this trend of rapid improvements will continue. How the MOSFET might continue to meet this expectation is the subject of this chapter. One overarching topic introduced in this chapter is the off-state current or the leakage current of the MOSFETs. This topic compliments the discourse on the on-state current presented in the previous chapter.

7.1 Technology Scaling—Small is Beautiful

Since the 1960's the price of one bit of semiconductor memory has dropped 100 million times and the trend continues. The cost of a logic gate has undergone a similarly dramatic drop. This rapid price drop has stimulated new applications and semiconductor devices have improved the ways people carry out just about all human activities. The primary engine that powered the ascent of electronics is "miniaturization". By making the transistors and the interconnects smaller, more circuits can be fabricated on each silicon wafer and therefore each circuit becomes cheaper. Miniaturization has also been instrumental in the improvements in speed and power consumption.

Gordon Moore made an empirical observation in the 1960's that the number of devices on a chip doubles every 18 months or so. The "Moore's Law" is a succinct description of the persistent periodic increase in the level of miniaturization. Each time the minimum line width is reduced, we say that a new technology generation or technology node is introduced. Examples of technology generations are 0.18 μm , 0.13 μm , 90nm, 65nm, 45nm...generations. The numbers refer to the minimum metal line width. Poly-Si gate length may be smaller. At each new node, the various feature sizes of circuit layout, such as the size of contact holes, are 70% of the previous node. This practice of periodic size reduction is called scaling. Historically, a new technology node is introduced every three years or so.

The main reward for introducing a new technology node is the reduction of circuit size by 2. (70% of previous line width means ~50% reduction in area, i.e. $0.7 \times 0.7 = 0.49$.) Since nearly twice as many circuits can be fabricated on each wafer with each new technology node, the cost per circuit is reduced significantly. That is the engine that drives down the cost of ICs.

Besides line width, some other parameters are also reduced with scaling such as the MOSFET gate oxide thickness and the power supply voltage. The reductions are chosen such that the transistor current density (I_{on}/W) increases with each new node. Also, the

smaller transistors and shorter interconnects lead to smaller capacitances. Together, these changes cause the circuit delays to drop (Eq. 6.7.1). Historically, integrated circuit speed has increased roughly 30% at each new technology node.

Scaling does another good thing. Eq. 6.7.6 shows that reducing capacitance and, especially, the power supply voltage is effective for lowering the power consumption. Thanks to the reduction in C and V_{dd} , power consumption per chip has increased only modestly per node in spite of the rise in switching frequency, f and (gasp) the doubling of transistors per chip at each technology node. If there had been no scaling, doing the job of a single PC microprocessor chip-- running 500M transistors at 2GHz using 1970 technology would require the electrical power output of a medium-size power generation plant.

In summary, scaling improves cost, speed, and power per function with every new technology generation. All of these attributes have been improved by 10 to 100 million times in four decades --- an engineering achievement unmatched in human history! When it comes to ICs, small is beautiful. Table 7.1 shows that scaling is expected to continue. But, what are the barriers to further scaling? Can scaling go on forever?

Table 7.1: Excerpt of 2003 ITRS technology scaling from 90nm to 22nm. The International Technology Roadmap for Semiconductors presents the industry's annually updated projection of future technologies and challenges [1]. HP:High Performance technology. LSTP: Low Standby Power technology for portable applications. EOT: Equivalent Oxide Thickness.

Year of Production	2004	2007	2010	2013	2016
Technology Node (nm)	90	65	45	32	22
HP physical Lg (nm)	37	25	18	13	9
EOT(nm) (HP/LSTP)	1.2/2.1	0.9/1.6	0.7/1.3	0.6/1.1	0.5/1.0
Vdd (HP/LSTP)	1.2/1.2	1.1/1.1	1.1/1.0	1.0/0.9	0.9/0.8
Ion/W,HP (mA/mm)	1100	1510	1900	2050	2400
Ioff/W,HP (mA/mm)	0.05	0.07	0.1	0.3	0.5
Ion/W,LSTP (mA/mm)	440	510	760	880	860
Ioff/W,LSTP (mA/mm)	1e-5	1e-5	6e-5	8e-5	1e-4

7.2 Subthreshold Current--- “Off” is not totally “Off”

Circuit speed improves with increasing I_{on} , therefore it would be desirable to use a small V_t . Can we set V_t at an arbitrarily small value, say 10mV? The answer is no.

At $V_{gs} < V_t$, an N-channel MOSFET is in the off-state. However, an undesirable leakage current can flow between the drain and the source. The MOSFET current observed at $V_{gs} < V_t$ is called the **subthreshold current**. This is the main contributor to the MOSFET off-state current, I_{off} . I_{off} is the I_d measured at $V_{gs}=0$ and $V_{ds}=V_{dd}$. It is important to keep I_{off} very small in order to minimize the static power that a circuit consumes even when it is in the standby mode. For example, if I_{off} is a modest 100nA per transistor, a cell-phone chip containing one hundred million transistors would consume so much standby current (10A) that the battery would be drained in minutes without receiving or transmitting any calls. A desk-top PC chip may be able to tolerate this static power but not much more before facing expensive problems with cooling the chip and the system.

Fig. 7-1 shows a typical subthreshold current plot. It is almost always plotted in a semilog I_{ds} versus V_{gs} graph. When V_{gs} is below V_t , I_{ds} is an exponential function of V_{gs} .

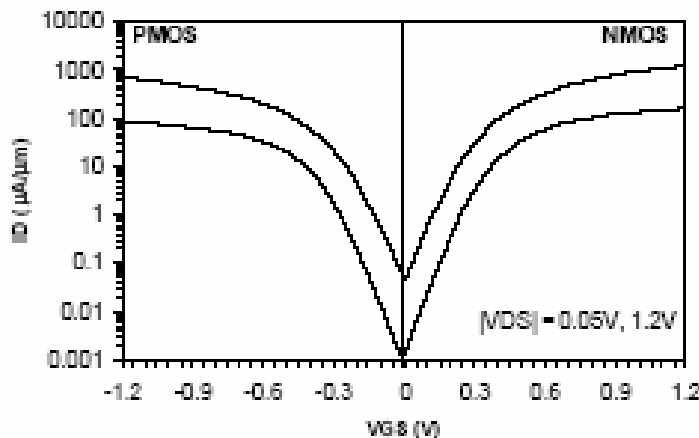


Figure 7-1 The current that flows at $V_{gs} < V_t$ is called the subthreshold current. $V_t \sim 0.2V$. The lower/upper curves are for $V_{ds}=50mV/1.2V$. After Ref. [2].

Fig. 7-2 explains the subthreshold current. At V_{gs} below V_t , the inversion electron concentration (n_s) is small but nonetheless can allow a small leakage current to flow between the source and the drain. In Fig. 7-2(a), a large V_{gs} would pull the E_c at the surface closer to E_t , causing n_s and I_{ds} to rise. From the equivalent circuit in Fig. 7-2(b), one can observe that

$$\frac{d\phi_s}{dV_{gs}} = \frac{C_{oxe}}{C_{oxe} + C_{dep}} \equiv \frac{1}{\eta} \quad (7.2.1)$$

$$\eta = 1 + \frac{C_{dep}}{C_{oxe}} \quad (7.2.2)$$

Integrating Eq. (7.2.1) yields

$$\phi_s = \text{constant} + V_g/\eta \quad (7.2.3)$$

I_{ds} is proportional to n_s , therefore

$$I_{ds} \propto n_s \propto e^{q\phi_s/kT} \propto e^{q(\text{constant} + V_g/\eta)/kT} \propto e^{qV_g/\eta kT} \quad (7.2.4)$$

The practical definition of V_t in experimental studies is the V_{gs} at which $I_{ds}=100\text{nA} \times W/L$. (Some companies may use 200nA instead of 100nA.)¹. Eq. (7.2.4) may be rewritten as

$$I_{ds}(nA) = 100 \cdot \frac{W}{L} \cdot e^{q(V_{gs}-V_t)/\eta kT} \quad (7.2.5)$$

Clearly, Eq. (7.2.5) agrees with the definition of V_t and Eq. (7.2.4). Recall that the function $\exp(qV_{gs}/kT)$ changes by 10 for every 60 mV change in V_{gs} , therefore $\exp(qV_{gs}/\eta kT)$ changes by 10 for every $\eta \times 60\text{mV}$. For example, if $\eta=1.5$, Eq. (7.2.5) states that I_{ds} drops by 10 times for every 90mV of decrease in V_{gs} below V_t . $\eta \times 60\text{mV}$ is called the **subthreshold swing** and represented by the symbol, S .

$$S = \eta \cdot 60mV \cdot \frac{T}{300} \quad (7.2.6)$$

$$I_{ds}(nA) = 100 \cdot \frac{W}{L} \cdot e^{q(V_{gs}-V_t)/\eta kT} = 100 \cdot \frac{W}{L} \cdot 10^{(V_{gs}-V_t)/S} \quad (7.2.7)$$

$$I_{off}(nA) = 100 \cdot \frac{W}{L} \cdot e^{-qV_t/\eta kT} = 100 \cdot \frac{W}{L} \cdot 10^{-V_t/S} \quad (7.2.8)$$

1. The alternative shown in How to Measure the V_t of a MOSFET in Sec 6.4 is not applicable at large V_{ds} .

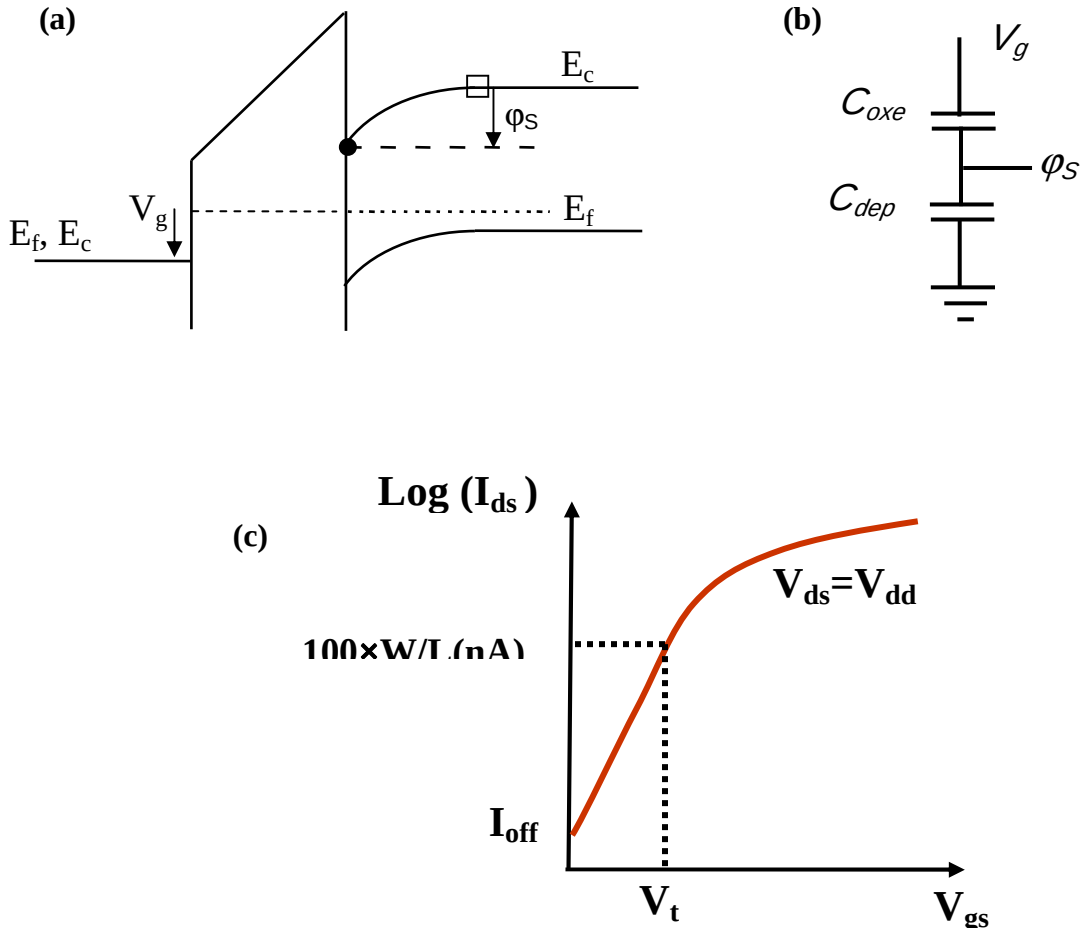


Figure 7-2: (a) When V_g is increased, E_c at the surface is pulled closer to E_f , causing n_s and I_{ds} to rise; (b) equivalent capacitance network; (c) Subthreshold IV with V_t and I_{off} .

For given W and L , there are two ways to minimize I_{off} illustrated in Fig. 7-2 (c). The first is to choose a large V_t . This is not desirable because a large V_t reduces I_{on} and therefore increases the gate delays (see Eq. (6.7.1)). The preferable way is to reduce the subthreshold swing. S can be reduced by reducing η . That can be done by increasing C_{oxe} (see Eq. 7.2.2), i.e. using a thinner T_{ox} , and by decreasing C_{dep} , i.e. increasing W_{dep} .² An additional way to reduce S , and therefore to reduce I_{off} , is to operate the transistors at a lower temperature. This last approach is valid in principal but rarely used because cooling adds considerable cost.

2. According to Eq. 6.5.2 and Eq. 7.2.2, η should be equal to m . In reality, η is larger than m because C_{oxe} is smaller at low V_{gs} (subthreshold condition) than in inversion due to a larger T_{inv} as shown in Fig. 5-25. Nonetheless, η and m are closely related.

Example: Subthreshold Leakage Current

An N-channel transistor has $V_t=0.34V$ and $S=85mV$, $W=10\mu m$ and $L=50nm$. A.) Estimate I_{off} . B.) Estimate I_{ds} at $V_g=0.17V$.

Answer:

A.) Use Eq. 7.2.6.

$$I_{off}(nA) = 100 \cdot \frac{W}{L} \cdot 10^{-V_t/S} = 100 \cdot \frac{10}{0.05} \cdot 10^{-0.34/0.085} = 2nA$$

B.) Use Eq. 7.2.7.

$$I_{ds} = 100 \cdot \frac{W}{L} \cdot 10^{(V_g-V_t)/S} = 100 \cdot \frac{10}{0.05} \cdot 10^{(0.17-0.34)/0.085} = 200nA$$

7.3 V_t Roll-off --- Short-channel MOSFETs are Hard to Turn Off

The previous section pointed out that V_t must not be set too low, otherwise I_{off} would be too large. The present section extends that analysis to show that the channel length (L) must not be too short. The reason is this: V_t drops with decreasing L as illustrated in Fig. 7-3. When V_t drops too much, I_{off} becomes too large and that channel length is not acceptable.

Sidebar: Gate Length (L_g) versus Channel Length (L) and Experimental Data versus Equations

Gate length is the physical length of the gate and can be accurately measured with a scanning electron microscope (SEM). It is carefully controlled in the fabrication plant (called **fab** in short). The channel length, in comparison, can not be determined accurately due to the lateral diffusion of the source and drain junctions. L tracks L_g well but the difference between the two just can not be quantified precisely. As a result, L_g is widely used in lieu of L in data collection and presentations such as in Fig. 7-3. L is used in theoretical equations but it is understood that L can not be known precisely for small real transistors. Thus we rely on measured data and complex computer simulations of

devices for precise device development and circuit design. On the other hand, we rely on the theoretical equations to guide the interpretation of the data, the design of new experiments, and the search for new innovative ideas.

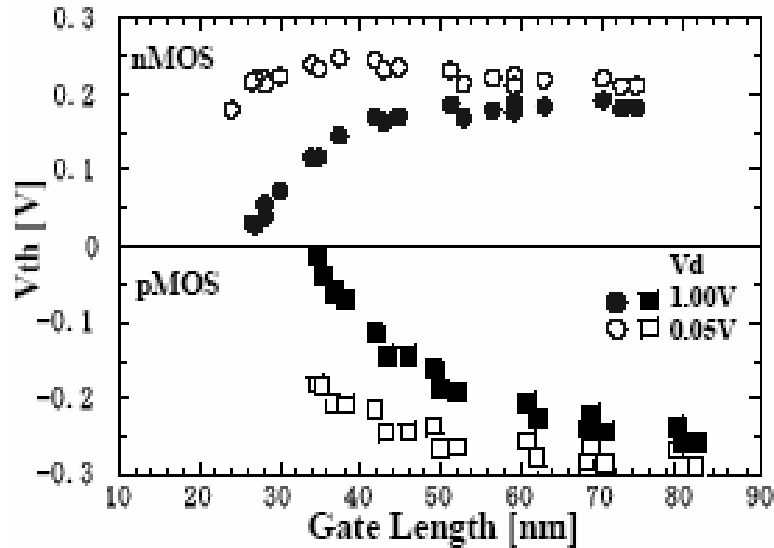


Figure 7-3 V_t decreases with decreasing L_g . This phenomenon is called V_t roll-off. It determines the minimum acceptable L_g because I_{off} is too large when V_t falls too low. After Ref. [3].

At certain L_g , V_t becomes so low that I_{off} becomes unacceptable (see Eq. 7.2.8). Device development engineers must design the device so that the V_t roll-off does not prevent the use of the targeted minimum- L_g , for example those listed in the third row of Table 7-1. Of course, lithography resolution must be able to support the L_g targets, too.

Why does V_t decrease with L ? Fig. 7-4 provides the answer. Fig. 7-4(a) shows the energy-band diagram along the semiconductor/insulator interface of a long channel device at $V_{gs}=0$. Fig. 7-4(b) shows the case at $V_{gs}=V_t$. In the case of (b), E_c in the channel is pulled lower than in case (a) and therefore is closer to the E_c in the source. When the channel E_c is only $\sim 0.2\text{eV}$ higher than the E_c in the source (which is also $\sim E_{fn}$), n_s in the channel reaches $\sim 10^{17}\text{ cm}^{-3}$ and inversion threshold is reached. We may say that a 0.2eV potential barrier is low enough to allow the electrons in the N^+ source to flow into the channel and then into the drain. The following analogy may be helpful for understanding the concept of the energy barrier height. The source is a reservoir of water;

the potential barrier is a dam; and V_{gs} controls the height of the dam. When V_{gs} is high enough, the dam is sufficiently low for the water to flow into the channel and the drain. That defines V_t .

Fig. 7-4(c) shows the case of a short-channel device at $V_{gs}=0$. If the channel is short enough, E_c will not be able to reach the same peak value as in Fig. 7-4(a). As a result, a smaller V_{gs} is needed in Fig. 7-4(d) than in Fig. 7-4(b) to pull the barrier down to 0.2eV. In other words, V_t is lower in case (d), the short channel device than in case (b), the long channel device. This explains the V_t roll-off shown in Fig. 7-3.

We can understand V_t roll-off from another approach. Fig. 7-5 shows a capacitor between the gate and the channel. It also shows a second capacitor, C_d , between the drain and the channel terminating at the location where E_c peaks in Fig. 7-4(d). As the channel length is reduced, the drain to source and drain to “channel” distance is reduced; therefore C_d increases. Do not be concerned with the exact definition or value of C_d . Just remember that it represents the strength of capacitive coupling in the complex two-dimensional structure of the drain and the channel.

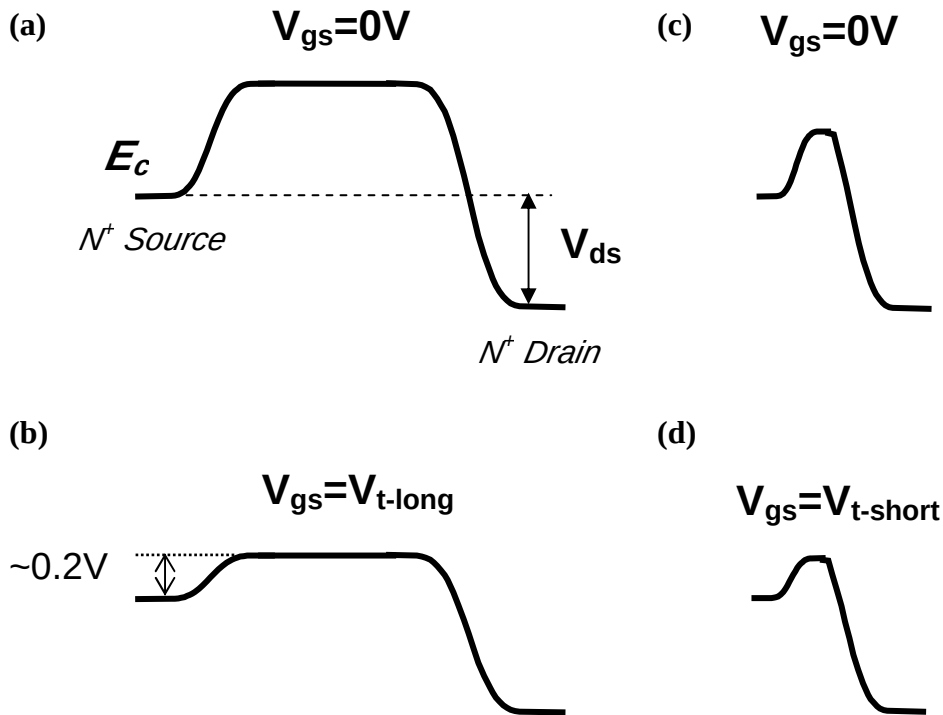


Figure 7-4 (a)-(d): Energy-band diagram from source to drain when $V_{gs}=0V$ and $V_{gs}=V_t$. (a)-(b) long channel; (c)-(d) short channel.

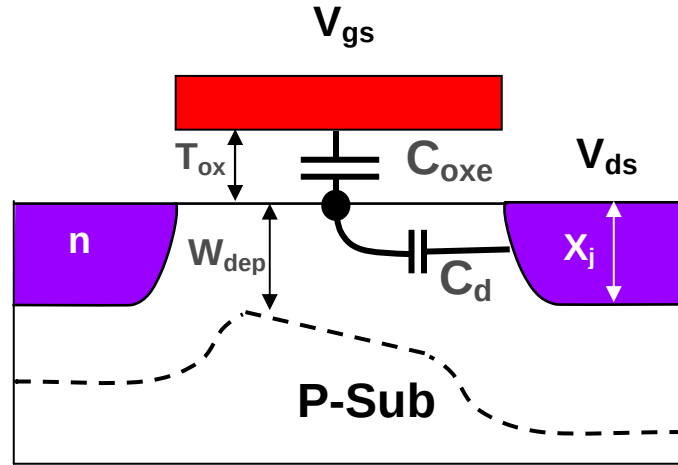


Figure 7-5 Schematic two-capacitor network in MOSFET. C_d models the electrostatic coupling between the channel and the drain. As the channel length is reduced, drain to “channel” distance is reduced; therefore C_d increases.

From this two-capacitor equivalent circuit, one immediately sees that the drain voltage has a similar effect as the gate voltage on the channel potential. V_{gs} and V_{ds} , together, determine the channel potential barrier height shown in Fig. 7-4. When V_{ds} is present, less V_{gs} is needed to pull the barrier down to $0.2eV$, therefore V_t is lower by definition. This understanding gives us a simple equation for V_t roll-off,

$$V_t = V_{t-long} - V_{ds} \cdot \frac{C_d}{C_{oxe}} \quad (7.3.1)$$

where V_{t-long} is the threshold voltage of a long-channel transistor, for which $C_d=0$. More exactly, V_{ds} should be supplemented with a constant that represent the effect of the built-in potentials between the N^- channel and the N^+ drain and source, about $0.4V$ [4].

$$V_t = V_{t-long} - (V_{ds} + 0.4) \cdot \frac{C_d}{C_{oxe}} \quad (7.3.2)$$

Using Fig. 7-5, one can intuitively see that as L decreases, C_d increases. Recall that the capacitance increases when the two electrodes are closer to each other. That intuition has been confirmed with 2-dimensioal computer simulations and analytical solutions of the Poisson equation. These analyses further indicate that C_d is an exponential function of L in this two-dimensional structure [5]. Therefore,

$$V_t = V_{t-long} - (V_{ds} + 0.4) \cdot e^{-L/l_d} , \quad (7.3.3)$$

$$\text{where } l_d \propto \sqrt[3]{T_{ox} W_{dep} X_j} \quad (7.3.4)$$

X_j is the drain junction depth. Eq. 7.3.3 provides a semi-quantitative model of the roll-off of V_t as a function of L and V_{ds} . At a very large L , V_t is equal to V_{t-long} as expected. The roll-off is an exponential function of L . The roll-off is also larger at larger V_{ds} , and the worst case is $V_{ds}=V_{dd}$. I_{off} becomes unacceptable when V_t is too small. This condition determines the minimum acceptable L . **The minimum acceptable L is several times of l_d .** In order to support the reduction of L at each new technology node, l_d must be reduced in proportion to L . This means that we must reduce T_{ox} , W_{dep} , and/or X_j . In reality all three are reduced at each node to achieve the desired reduction in l_d . Reducing T_{ox} increases the gate control or C_{oxe} . Reducing X_j decreases C_d by reducing the size of the drain electrode. Reducing W_{dep} also reduces C_d by introducing a ground plane (the neutral region of the substrate or the bottom of the depletion region) that shields the channel from the drain.

One way to summarize the message of Eq. 7.3.4 is that **vertical dimensions in a MOSFET (T_{ox} , W_{dep} , X_j) must be reduced in order to support the reduction of gate length.**

7.4 Reducing the Gate Insulator Thickness and T_{oxe}

SiO_2 has been the preferred gate insulator for silicon MOSFET since its very beginning in the 1960's and the oxide thickness has been reduced over the years from 300nm for 10 μm technology to 1.2nm for 65nm technology. There are two reasons for the relentless drive to reduce the oxide thickness. First, a thinner oxide, i.e. a larger C_{ox} raises I_{on} . A large I_{on} is desirable for maximizing the circuit speed (see Eq. 6.7.1). The second reason is to control V_t roll-off (and therefore the subthreshold leakage) in the presence of falling L according to Eqs. 7.3.3 and 7.3.4. One must not underestimate the importance of the second reason. Fig. 7-6 shows that the oxide thickness has been scaled roughly in proportion to the line width.

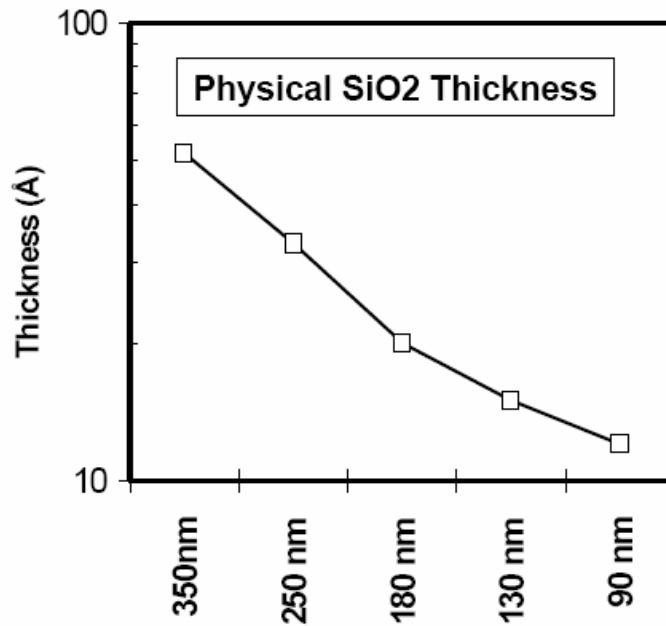


Figure 7-6: Oxide thickness has been scales roughly in proportion to the line width.

So, thinner oxide is desirable. What, then, prevents engineers from using arbitrarily thin gate oxide films? Manufacturing thin oxide is not easy, but as Fig. 6-5 illustrates, it is possible to grow very thin and uniform gate oxide films with high yield. Oxide breakdown is another limiting factor. If the oxide is too thin, the electric field in the oxide can be so high as to cause destructive breakdown. (See the sidebar: SiO₂ Breakdown Electric Field.) Yet another limiting factor is that long term operation at high field, especially at elevated chip operating temperatures, breaks the weaker atomic bonds at the Si/SiO₂ interface thus creating oxide charge and V_t shift (see Sec. 5.7). V_t shifts cause circuit behaviors to change and raise reliability concerns.

For SiO₂ films thinner than 1.5nm, tunneling leakage current becomes the most serious limiting factor. Fig. 7-7(a) illustrates the phenomenon of tunneling. Fig. 7-7(b) shows that the very rapid rise of the SiO₂ leakage current with decreasing thickness agrees with the tunneling model prediction [6]. At 1.2nm, SiO₂ leaks 10^3 A/cm². If an IC chip contains 1mm² total area of this thin dielectric, the chip oxide leakage current would be 10A. This large leakage would drain the battery of a cell phone in minutes.

Researchers are developing high-k dielectrics to replace SiO₂. For example, HfO₂ has a relative dielectric constant (k) of ~24, six times large than that of SiO₂. A 6nm thick HfO₂ film is equivalent to 1nm thick SiO₂ in the sense that both films produce the same C_{ox} . We say that this HfO₂ film has an **equivalent oxide thickness** or **EOT** of 1nm. However, the HfO₂ film presents a much thicker (albeit a lower) tunneling barrier to the electrons and holes. The consequence is a leakage current that is several orders of magnitude smaller than that in SiO₂ as shown in Fig. 7-7(b) and (c). A metal gate is used

to reduce the poly-Si gate depletion and EOT in 7-7(c) [7]. Other candidates of high-k gate dielectric include ZrO_2 and Al_2O_3 . The difficulties of adopting high-k dielectrics in IC manufacture include chemical reactions between them and the silicon substrate and gate, lower surface mobility than the Si/SiO₂ system. These problems can be minimized by inserting a thin SiO₂ interfacial layer between the silicon substrate and the high-k dielectric and using a metal gate instead of a poly-Si gate.

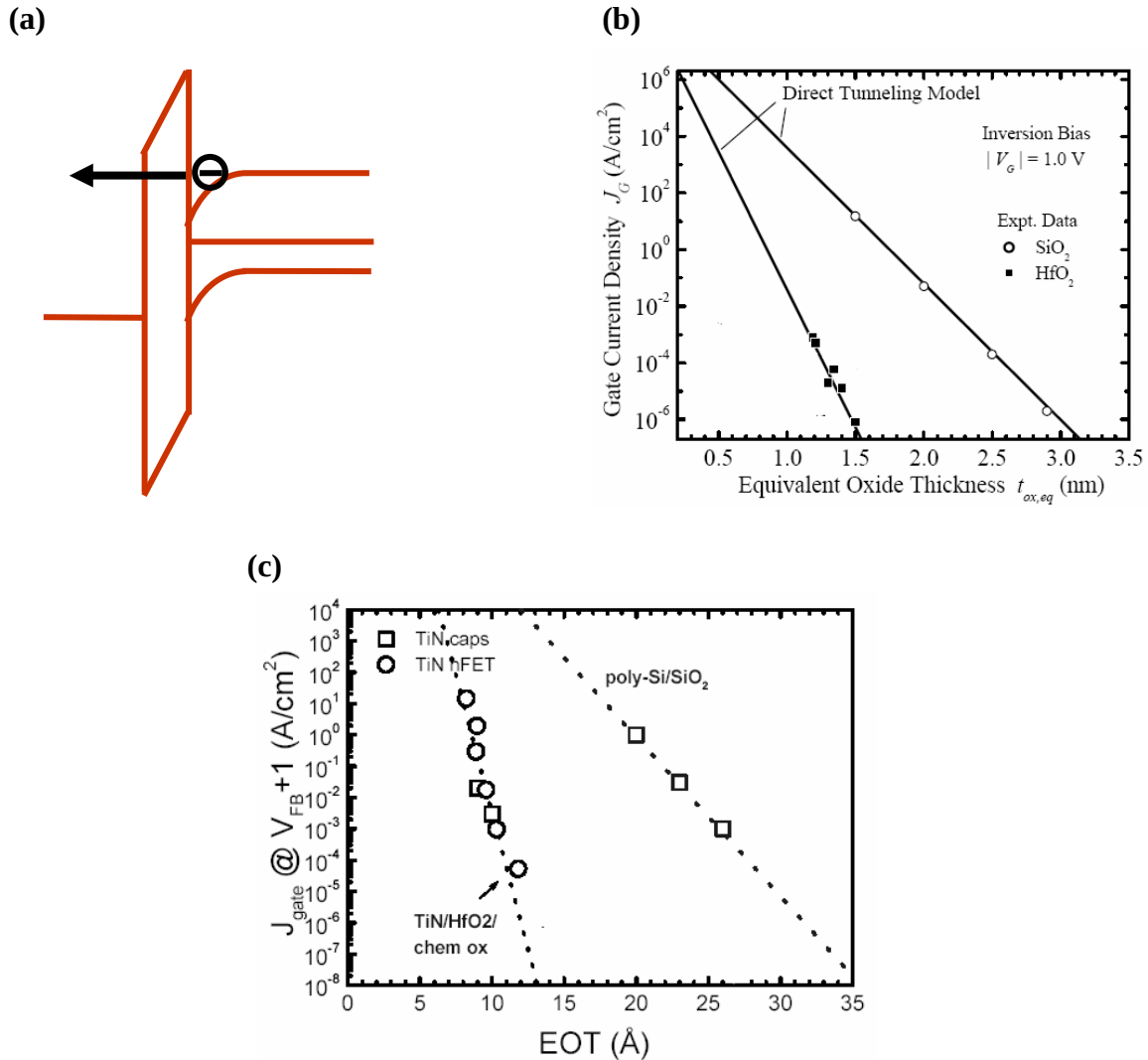


Figure 7-7 (a)-(c): (a) Energy band diagram in inversion showing electron tunneling path through the gate oxide. 1.2 nm SiO₂ conducts 10^3 A/cm² of leakage current. High-k dielectric such as HfO₂ has several orders lower leakage; After Ref. [6] (c) HfO₂ is used with a TiN metal gate and a thin SiO₂ at the Si interface produced by wet chemistry. After Ref. [7]

Note that Eq. 7.3.4 contains the electrical oxide thickness, T_{oxe} , defined in Eq. 5.9.2. Besides T_{ox} or EOT in the case of high-k dielectric, poly-Si gate depletion layer also needs to be minimized. A metal gate would be the ultimate gate material in this respect. The challenge there is to find metals that have work functions close to those of N^+ and P^+ poly-Si.

In addition, T_{inv} needs to be minimized. The material parameters that determine T_{inv} is the electron and hole effective masses. A larger effective mass leads to a thinner T_{inv} . Unfortunately, a larger effective mass leads to a lower mobility. Fortunately, the effective mass is a function of the spatial direction of carrier motion in a crystal. The effective mass in the direction normal to the channel determines T_{inv} , while the effective mass in the plane of the channel determines the surface mobility, μ_s . It may be possible to choose a semiconductor and a wafer surface orientation (see Fig. 1-2) that together produce large m_n and m_p normal to the channel and small m_n and m_p in the plane of the channel.

SiO_2 –Breakdown Electric Field

What is the breakdown field of SiO_2 ? There is no one simple answer because the breakdown field is a function of the stress time. If a one second (1s) voltage pulse is applied to a 10nm SiO_2 film, 15V is needed to breakdown the film for a breakdown field of 15MV/cm. The breakdown field is significantly lower if the same oxide is tested for one hour. The field is lower still if it is tested for one month. This phenomenon is called **time-dependent dielectric breakdown**. Many IC applications require a device lifetime of 10 years. Clearly, manufacturers can not afford the time to actually measure the 10 yr breakdown fields for new oxide technologies. Instead, researchers have predicted the 10 yr breakdown fields based on short-term tests in combination with theoretical models of the physics of oxide breakdown. In retrospect, the most optimistic of the predictions, 7MV/cm for 10year operation, was basically right, and SiO_2 thickness has been scaled further than the other models predicted possible [8].

This breakdown model suggests that carrier tunneling at high field introduces holes into SiO_2 . Holes cause the break-up of the weaker Si-O bonds in amorphous SiO_2 thus creating oxide defects. This process progresses more rapidly at random spots in the oxide sample where the densities of the weaker bonds happen to be statistically high. When the generated defects reach a critical density at any one spot, breakdown occurs. In a longer term stress test, the breakdown field is lower because a lower rate of defect generation is sufficient to build up the critical defect density over the longer stress time. A fortuitous fact is that the breakdown field increases somewhat with decreasing oxide thickness. The reason is that a larger fraction of holes may escape the thinner film without generating defects; therefore a higher field can be tolerated.

7.5 How to Reduce W_{dep}

Eq. 7.3.4 suggests that a small W_{dep} helps to control V_t roll-off and enable the use of a shorter L . W_{dep} can be reduced by increasing the substrate doping concentration, N_{sub} because W_{dep} is proportional to $1/\sqrt{N_{sub}}$. However Eq. 5.4.3, repeated here,

$$V_t = V_{fb} + 2\phi_B + \frac{\sqrt{qN_{sub}2\epsilon_s2\phi_B}}{C_{ox}} \quad 7.5.1$$

tells us that, if V_t is not to increase, N_{sub} must not be increased unless C_{ox} is increased, i.e. T_{ox} is reduced. It can be shown that W_{dep} can only be reduced in proportion to T_{ox} .

$$V_t = V_{fb} + 2\phi_B + \frac{2\epsilon_s2\phi_B}{C_{ox}W_{dep}} \quad 7.5.2$$

This fact further highlights the importance of reducing T_{ox} as the main enabler of L reduction according to Eq. 7.3.4.

There is another way of reducing W_{dep} --- adopt the steep retrograde doping profile illustrated in Fig. 6-12. In this case, W_{dep} is determined by the thickness of the lightly doped surface layer. It can be shown that V_t of an MOSFET with ideal retrograde doping is

$$V_t = \frac{E_g}{q} - 0.1 + \left(\frac{E_g}{q} - 0.1 \right) \frac{\epsilon_{si}T_{ox}}{\epsilon_{oxi}L_{rg}} \quad 7.5.3$$

where L_{rg} is the thickness of the lightly doped thin layer. The derivation of Eq. 7.5.3 is left as an exercise for the interested readers in the Problems at the end of the chapter. Again, W_{dep} ($=L_{rg}$) can only be scaled in proportion to T_{ox} if V_t is to be kept constant. However, W_{dep} in an ideal retrograde device can be about half the X_{dep} of a uniformly doped device and yield the same V_t . That is an advantage of the retrograde doping. Another advantage of retrograde doping is that ionized impurity scattering (see Sec. 2.2.2) in the inversion layer can be reduced and surface mobility can be higher. However, dopant diffusion makes it difficult to fabricate a retrograde profile with a very thin lightly doped layer, i.e. a very small W_{dep} unless process temperature is further lowered.

Predicting the Ultimate Low Limit of Channel Length – A Retrospective

Assuming that lithography and etching technologies can produce as small features as one desires, what is ultimate lower limit of MOSFET channel length? When the channel length is too small, it would have too large an I_{off} and ceases to be a good transistor for practical purposes. What is the ultimate limit of the channel length?

In the 1970's the consensus in the semiconductor industry was that the ultimate lower limit of channel length is 500nm. In the 80's, the consensus was 250nm. In the 90's, it was 100nm. Now it is shorter than 10nm. What made the most knowledgeable experts in the industry and universities underestimate how far channel length can be scaled?

A review of the historical literature reveals that the researchers were mistaken about the lower limit of gate oxide thickness. In the 70's it was thought ~15nm would be the limit. In the 80's, it was 8nm, and so on. Since the T_{ox} estimate was off, the estimates of the minimum acceptable W_{dep} and therefore the minimum L would be off according to Eq. 7.3.4.

Here is an intriguing note about W_{dep} scaling. A higher N_{sub} in Eq. 7.5.1 (and therefore a smaller W_{dep}) is allowable if V_t is allowed to be larger. This larger V_t can be brought back down with a body (or well) to source bias voltage, V_{bs} . The required V_{bs} is a forward bias across the body-source junction. The forward bias is acceptable, i.e. the forward bias current is small as long as V_{bs} is kept below 0.6V.

7.6 Shallow Junction Technology

Fig. 7-8, first introduced as Fig. 6-24(b), shows the cross-sectional view of a typical drain (and source) junction. Extra process steps are taken to produce the **shallow junction extension** between the deep N^+ junction and the channel. This shallow junction is needed because the drain junction depth must be kept small according to Eq. 7.3.4. In order to keep this junction shallow, only short annealing at the lowest necessary temperature is used to activate the dopants and anneal out the implantation damages in the crystal. Because dopant diffusion can not be totally avoided, the doping concentration in the shallow junction extension must be kept low (much lower than the N^+ doping density). Shallow junction and light doping combine to produce an undesirable parasitic resistance that reduces the precious I_{on} . That is a price to pay for suppressing V_t roll-off and the subthreshold leakage current. Farther away from the channel, as shown in Fig. 7-8, a deeper N^+ junction is used to minimize total parasitic resistance. However, even the depth of the N^+ junction should be kept shallow to help V_t roll-off.

One possible way to beat the tradeoff between the junction depth and low series resistance is to replace the shallow junction extension with a thin layer of metal or silicide. This is theoretically possible but the metal or silicide must be chosen such that there is not a large energy barrier (see Ch. 9) between it and the silicon channel and not a large leakage between it and the substrate [9].

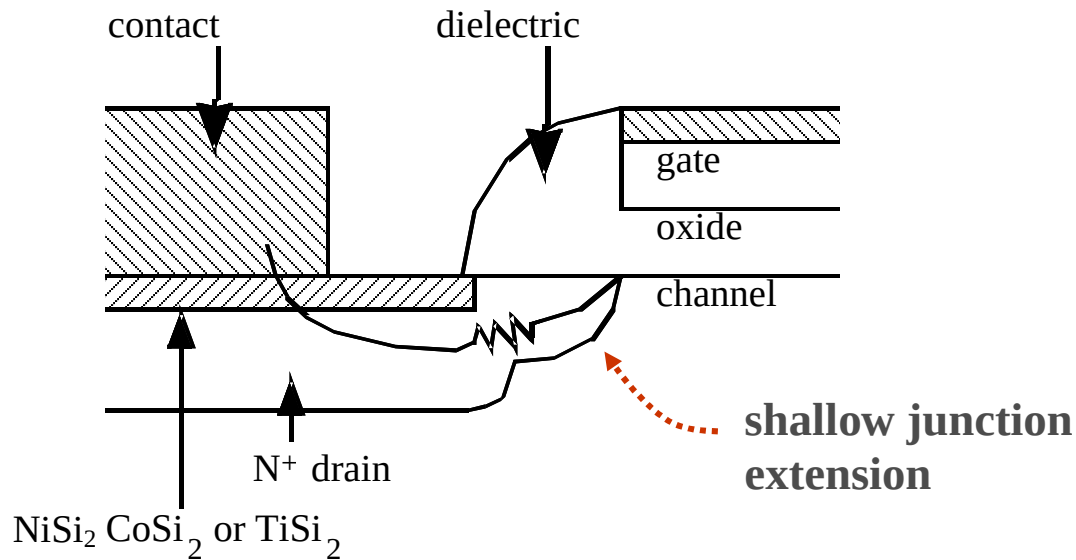


Fig. 7-8 Cross-sectional view of a MOSFET drain junction. The shallow junction extension next to the channel helps to suppress the V_t roll-off.

7.7 Trade-off between I_{on} and I_{off}

I_{off} would not be a problem if V_t is set at a very high value. That is not acceptable because a high V_t would reduce I_{on} and therefore reduce circuit speed. Using a larger V_{dd} can raise I_{on} , but that is not an acceptable solution because a larger V_{dd} would raise the power consumption, which is already too large for comfort. Most other changes that could reduce the leakage would also hurt I_{on} .

The salient exception is to use a smaller T_{ox} . That improves both I_{on} and V_t roll-off. Unfortunately, even T_{ox} reduction is no longer a cure without a serious side effect. In fact, the side effect--large dielectric tunneling leakage--has made SiO_2 thickness reduction beyond 1nm more harmful than helpful.

Question: Does any of the following changes contribute to both leakage reduction and I_{on} enhancement? A larger V_t . A larger L . A shallower junction. A smaller V_{dd} (hint: the worst case V_{ds} in Eq. 7.3.3 is V_{dd}).

Fig. 7-9 shows a plot of $\text{Log } I_{\text{off}}$ versus I_{on} [2]. The trade-off between the two is clear. Higher I_{on} goes hand-in-hand with larger I_{off} . The spread in I_{on} (and I_{off}) is due to a combination of unintentional manufacturing variances in L_g and intentional difference in drawn gate length.

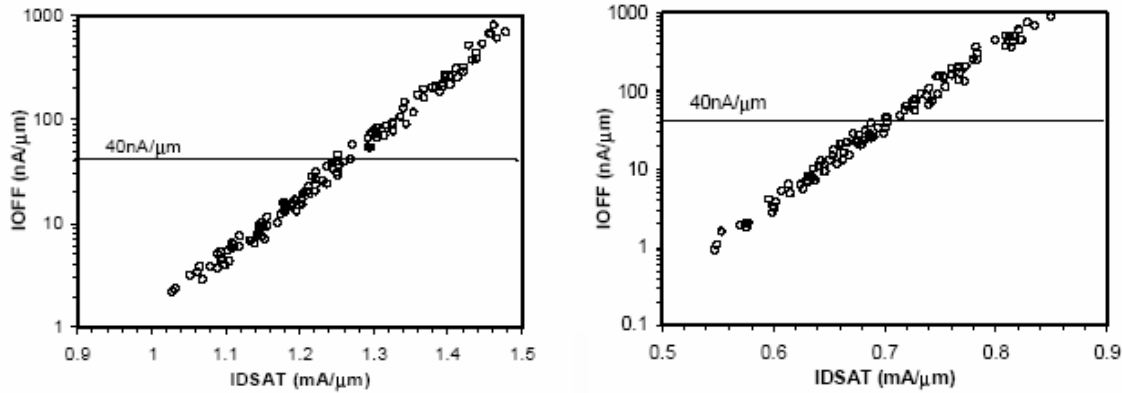


Figure 7-9 $\text{Log } I_{\text{off}}$ versus I_{on} . The spread in I_{on} (and I_{off}) is due to a combination of intentional differences and unintentional variances in L_g . After Ref. [2]

There are several techniques at the border between device technology and circuit design that can help to relax the conflict between I_{on} and leakage. In a large circuit such as a microprocessor, only some circuit blocks need to operate at high speed at a given time and other circuit blocks operate at lower speed or are idle. V_t can be set relatively low to produce large I_{on} so that circuits that need to operate at high speed can do so. A substrate or well bias voltage, V_{sb} in Eq. 6.4.6, is applied to the other circuit blocks to raise the V_t and suppress the subthreshold leakage. This technique requires intelligent control circuits to apply V_{sb} where and when needed. This technique is practical and often used. It also provides a way to compensate for the chip-to-chip and block-to-block variations in V_t that results from non-uniformity among devices due to imperfect manufacturing equipment and process. An interesting alternative is to apply a forward source-body bias to reduce V_t when and where high speed is needed. If the forward bias is lower than 0.6V, the diode forward current is acceptable due to the small junction area. The advantage of this alternative is that W_{dep} is reduced by the forward bias and V_t roll-off is improved (see Eq. 7.3.4).

Another technique gives circuit designers two or three (or even more) V_t to choose. A large circuit may be designed with only the high- V_t devices first. Circuit timing simulations are performed to identify those signal paths and circuits where speed must be tuned up. Intermediate- V_t devices are substituted into them. Finally, low- V_t devices are substituted into those few circuits that need even more help with speed. A similar strategy provides multiple V_{dd} rather than multiple V_t . A higher V_{dd} is provided to a